APL LOGISTICS DELIVERY DASHBOARD

Power BI Project Report

Delivery Performance • Delay Patterns • Operational Exploration

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Project type	Interactive business intelligence dashboard
Tool	Microsoft Power BI Desktop

This is a project report for documentation/submission. It is not represented as a peer-reviewed or published research paper.

Abstract

This project presents an interactive Power BI dashboard for exploring logistics delivery performance. It focuses on delivery timing, delivery status, shipping mode comparisons, and regional variation. KPI measures and interactive filters allow users to examine patterns across selected markets, shipping modes, customer segments, departments, and delivery statuses.

The dashboard supports descriptive exploration of the available dataset; it does not independently establish causal relationships or validate operational explanations for delays.

1. Introduction

Delivery performance is an important operational dimension in logistics. Reviewing delivery timing and status alongside shipping modes and regional groupings can help identify patterns that may warrant further investigation.

The APL Logistics Delivery Dashboard consolidates selected indicators into an interactive Power BI report, allowing users to filter and compare metrics rather than relying only on static tables.

2. Project Objectives

• Monitor delivery performance using summary indicators such as average delivery delay and on-time delivery rate.
• Explore late-delivery patterns using the dashboard's risk measure.
• Compare shipping modes using the efficiency indicator and scheduled-versus-actual shipping duration.
• Examine regional variation through the Regional Delay Index.
• Enable filtering by market, shipping mode, customer segment, department, and delivery status.

3. Dashboard Metrics

The dashboard includes the following KPI measures:

Metric	Purpose
Average Delivery Delay	Summarizes average delivery delay for the current filter selection.
Late Delivery Risk Ratio	Displays the implemented late-delivery risk measure.
Shipping Mode Efficiency Index	Provides a comparative indicator for shipping mode analysis.
Regional Delay Index	Supports comparison of delay patterns across markets/regions.
On-Time Delivery Rate	Summarizes the on-time delivery rate according to the dashboard's definition.

4. Visualizations and Interactivity

Delivery status: a donut chart displays the distribution of delivery statuses.
Shipping mode analysis: visuals support comparison of shipping mode efficiency and scheduled versus actual shipping duration.
Regional comparison: a bar chart compares the Regional Delay Index across markets.
Slicers: Market, Shipping Mode, Customer Segment, Department Name, and Delivery Status provide interactive filtering. KPI cards and visuals respond to filter context where configured.

5. Methodology

Data preparation: Power Query within Power BI is used for data preparation and transformation.
Metric development: DAX measures calculate and display dashboard indicators.
Visualization: KPI cards, charts, and slicers are arranged in an interactive report.
Exploration: users apply filters and compare resulting values and patterns. Interpretation should consider dataset

coverage, metric definitions, and data quality.

6. How to Use the Dashboard

1. Open the PBIX file in Microsoft Power BI Desktop.
2. If prompted, locate or update the source data connection.
3. Refresh the data if necessary and confirm that visuals load.
4. Use slicers to select a market, shipping mode, customer segment, department, or delivery status.
5. Review changes in KPI values and charts; clear filters to return to the overall view.

7. Findings and Interpretation

The dashboard is structured to help users identify where delivery measures differ across selected categories. This report does not claim that any specific market, shipping mode, or segment performs better or worse because verified numeric results and validated business conclusions have not been included here.

A formal research study should document dataset provenance, precise metric definitions, data-quality checks, appropriate statistical analysis, and validation with domain stakeholders.

8. Limitations

The dashboard depends on the completeness, accuracy, and definitions of the underlying dataset. KPI names alone do not specify formulas; users should consult the implemented DAX measures and data dictionary. The dashboard is descriptive and does not independently establish why a delay occurred or whether an operational factor caused it.

9. Conclusion

The APL Logistics Delivery Dashboard demonstrates how Power BI can organize logistics performance indicators into an interactive report. It combines KPI summaries, delivery-status and shipping-mode visuals, regional comparisons, and slicers to support exploratory analysis. The project provides practical experience in data preparation, DAX measures, and dashboard design.

Project Information

Project: APL Logistics Delivery Dashboard
Developer: Priya Jangra
Platform: Microsoft Power BI Desktop
Deliverables: PBIX dashboard and project report

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